Thick Cloud Removal of Landsat Time-Series Using Convolutional LSTM With Embedded Residual Modules

Lanxing Wang, Qunming Wang, and Peter M. Atkinson

***Abstract*— Extensive cloud contamination severely hinders the interpretation of optical remote sensing images. Existing cloud removal methods focus primarily on the reconstruction of individual cloudy images, with few studies addressing the reconstruction of cloudy time-series images. Furthermore, current methods tend to prioritize using cloud-free auxiliary images while overlooking valuable information present in the cloudy auxiliary images that are temporally closer to the target cloudy image. In this article, we proposed a deep network called Res-cLSTM to reconstruct cloudy time-series images. Res-cLSTM processes time-series images sequentially using convolutional LSTM, synthesizing long- and short-term memory streams to match the complex temporal relationships amongst them. Then, Res-cLSTM further decodes the feature maps using a refined residual module with skip connections, resulting in the final output. Simulated and real cloud removal experiments on Landsat 8 OLI time-series data across five different regions demonstrated that Res-cLSTM is an effective cloud removal method, which can produce more accurate predictions than three benchmark approaches. For example, for reconstruction of the cloudy time-series of three simulated cloudy areas, the average correlation coefficient (CC) of the Res-cLSTM prediction is about 0.01, 0.04, and 0.04 larger than that of the second most accurate method [i.e., autoencoder (AE)]. As a lightweight network, Res-cLSTM does not require global sampling of training data and can fully exploit the valuable information in the noncloud regions of cloudy time-series images to facilitate cloud removal. Moreover, Res-cLSTM demonstrates robustness to thin cloud omission and exhibits a faster convergence rate and, thus, holds great potential for practical applications requiring real-time processing.**

***Index Terms*— Cloud removal, deep learning, Landsat 8, thick clouds, time-series.**

I. Introduction

OPTICAL remote sensing images provide important data support for continuous land surface monitoring [1], [2], [3], [4] and provide unprecedented opportunities for exploring natural [5], [6], [7], [8] and anthropogenically affected processes [9], [10], [11], [12], [13]. However, the number of effective optical satellite sensor images available is considerably less than the number of images measured according to the satellite's stated nominal revisit frequency. The main reason for the data gap is that clouds (and their shadows) obscure the observation scene. Studies have shown that clouds obscure an average of 67% of the Earth's surface at any one time [14]. Cloud detection is often used as a preprocess to identify cloud regions in the image, commonly leading to the generation of cloud masks to aid subsequent reconstruction [15], [16], [17], [18], [19]. In reality, clouds vary in thickness, from thin to thick. Thin cloud can be removed with reference to partially visible information under the cloud [20], [21], [22], [23], [24]. Thick clouds, however, block the propagation of light across all bands, resulting in a complete loss of land surface information and a greater challenge to recover the information below. This article focuses on thick cloud removal and reconstruction of the information under thick clouds based on defined cloud masks.

Due to the mobility of clouds, images acquired on other dates at the same location may contain valid information about the target cloud region, and such images are called auxiliary images. Current mainstream methods for cloud removal often employ temporally auxiliary images, which can be categorized into two groups: those based on a single auxiliary image and those based on multiple temporal auxiliary images [25], [26], [27], [28]. The two groups of methods can be further classified into methods utilizing homologous auxiliary data and methods employing heterogeneous auxiliary data. Given the temporal dependence as well as temporal variation, the auxiliary images need to be sufficiently close to the target cloudy image temporally.

The single auxiliary image-based methods construct the mapping relationship between the auxiliary image and the target cloudy image based on the common noncloud data between them and utilize the valid data in the auxiliary image to reconstruct the target cloud region [29]. The auxiliary data can be classified into homologous and heterologous sources. The homologous data refer to the temporally close data from the same sensor or satellite series. Considering the complexity of the mapping relationship between the auxiliary image and the target cloudy image, machine learning-based methods have been increasingly investigated for cloud removal [30]. For example, Melak et al. [31] employed an autoencoder (AE) neural network to remove clouds, which consists of fully connected layers and ReLU functions, for training and prediction based on individual pixels or

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patches. Gao et al. [32] proposed a deep code regression (DCR) model, combined with an autoregressive structure, to reconstruct the target cloud region. In [33], an auxiliary image was initially employed to derive preliminary predictions of the target cloud region using linear regression, which was subsequently refined using a convolutional neural network (CNN) to achieve the final predictions. Tao et al. [34] sorted the training samples based on texture complexity and used self-paced learning to train a generative adversarial network for cloud removal. The aforementioned methods based on homologous auxiliary image capitalize on the spectral consistency. However, frequent cloud occlusion and inherent satellite revisit periods often render the available homologous auxiliary images temporally distant from the target cloudy image. Consequently, researchers have explored the use of synthetic aperture radar (SAR) images, which can penetrate clouds, as a source of auxiliary images (i.e., heterologous auxiliary data) [35], [36], [37], [38], [39]. Deep learning methods have demonstrated competent performance in establishing cross-spectral mapping between SAR and optical images. For example, Xiang et al. [40] proposed a two-step cloud removal method that first converts the SAR image into the corresponding optical image and subsequently refines the cloud region of the optical image using a cloud-guided fusion network to achieve the final predictions. For methods that utilize only SAR images as auxiliary information, it remains a great challenge to mitigate the effects of severe speckle noise while effectively extracting relevant information from the original data [41], [42]. Overall, single auxiliary image-based methods are generally simple to implement as they do not necessitate the collection of extensive training datasets.

Methods based on multitemporal auxiliary images also involve the use of either homologous or heterologous auxiliary images. Zhu et al. [43] developed a reconstruction model by using the Julian day of the time-series acquisition date as the independent variables and the corresponding pixel values as the dependent variable, enabling the reconstruction of cloud pixels for any given date. To characterize the complex mapping relationship between the time-series, deep learning-based methods have also been developed. Conventional methods focus on fitting explicit relationships based on known information to construct prediction models, making them difficult to handle complex mapping relationships in large-scale cloud-contaminated scenes and time-series data [44], [45]. Therefore, deep learning-based methods have been developed rapidly. Chen et al. [46] developed a spatiotemporal information-based neural network known as STnet, which effectively reconstructs a target cloudy image by leveraging spatiotemporal feature fusion modules for spatiotemporal information learning. In addition, deep learning techniques have also been used as postprocessing to further optimize the results of tensor completion algorithms. Zhang et al. [47] combined model- and data-driven approaches by employing third-order tensor singular value decomposition along with 3-D CNNs to reconstruct time-series cloudy images. Zheng et al. [48] introduced tensor network decomposition and integrated the initial known mask iteration into the optimization process to refine the cloud mask further while using multitemporal data for cloudy image reconstruction. With respect to the use of heterologous auxiliary data, the powerful learning ability of deep learning models facilitates the fusion of SAR and optical auxiliary images [49]. Specifically, time-series optical images and SAR images can be explored simultaneously for cloud removal of temporally neighboring cloudy images [49]. For cloud removal from Landsat series data, Li et al. [50] first applied a spatiotemporal nonlocal filtering model to fuse a homologous auxiliary image with coarse-resolution time-series images (i.e., MODIS) to obtain a cloud-free image. Then, the cloud-contaminated areas in the target cloudy image were removed using the cloud-free image through nonnegative matrix factorization.

Despite the above progress, existing cloud removal methods exhibit several limitations. Specifically, single auxiliary image-based methods typically do not require a large number of training data but often necessitate the use of cloud-free auxiliary images [29], [31], [32]. For the methods based on single auxiliary images, the accuracy of the final prediction hinges on the validity of the cloud-free auxiliary images temporally closest to the target cloudy image [30]. However, clouds remain prevalent. Due to the limited temporal resolution of satellite sensors, there can be substantial time gaps (sometimes exceeding a year) between the available cloud-free auxiliary images and the target cloudy images. In such cases, a cloudy auxiliary image that is temporally closer to the target cloudy image may provide more valuable information. Moreover, since the cloud-contaminated areas in the auxiliary image are likely to overlap with the target cloud regions, utilizing multitemporal cloudy auxiliary images could be an effective approach. However, cloud removal methods that rely on time-series images typically require extensive data for model training, creating high demands for both the quantity and quality of training datasets. Without sufficient training data, the models may struggle with generalization [51], [52], [53]. Additionally, current research in cloud removal focuses largely on reconstructing individual cloudy images, with comparatively few methods addressing the reconstruction of time-series cloudy images jointly. Yet, time-series data contain critical information regarding temporal changes [54]. By reconstructing time-series images, dynamic phenomena such as landscape alterations, plant growth, and urban expansion can be more comprehensively monitored and analyzed [55], [56], which cannot be achieved using a single remote sensing image.

In light of the limitations discussed, this article proposed a residual block-enhanced convolutional long short-term memory network (Res-cLSTM) for reconstructing time-series cloudy images. Originally designed for precipitation prediction, convolutional LSTM [57] excels at capturing complex temporal relationships within time-series data. Res-cLSTM operates by sequentially inputting cloudy time-series images into the convolutional LSTM module, which decodes them while fitting the intricate temporal change relationships based on both long-term and short-term memory flows. In the Res-cLSTM model, convolution operations are employed to integrate spatial information. Moreover, the short-

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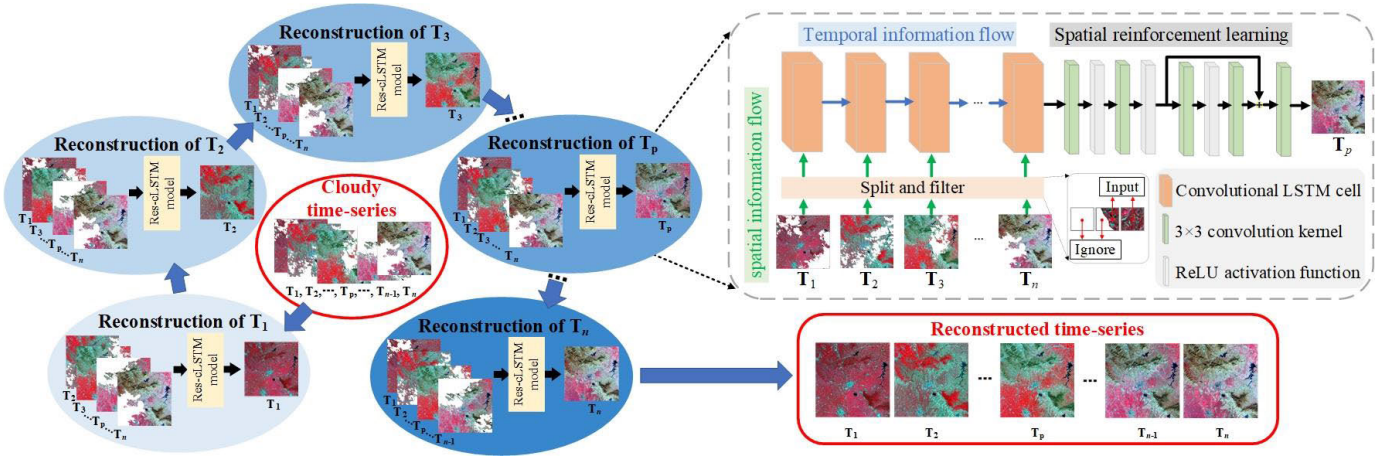


Fig. 1. Overview of the proposed Res-cLSTM.

circuit connections of the residual module can enhance the generalization ability, reduce the risk of overfitting, and stabilize the optimization process [58]. Taking this into consideration, Res-cLSTM then uses a residual module with a short-circuit connection to further refine the description of the feature map and output the final result.

When using cloudy auxiliary images, the cloud areas may overlap with the target cloud areas. In such cases, methods based on single auxiliary image need to combine reconstruction results from multiple auxiliary images in turn to achieve a complete prediction. This accumulates the uncertainty and computational cost step by step. Methods that utilize time-series auxiliary images typically input the data simultaneously into the network, limiting their ability to fully learn the inherent temporal variations. In contrast, Res-cLSTM processes time-series auxiliary images sequentially in chronological order, thereby enhancing its ability to learn temporal variations. Moreover, when time-series are cloudy, the integration of long- and short-term memory in Res-cLSTM effectively facilitates the complementarity of valuable information in noncloud areas across the time-series.

In summary, the Res-cLSTM network offers several advantages.

1) Different from traditional cloud removal methods that focus solely on reconstructing individual cloudy images, Res-cLSTM aims to reconstruct the entire cloudy time-series.
2) Res-cLSTM is a lightweight network with a relatively small number of parameters, allowing it to be trained solely on noncloud data from cloudy time-series images. Thus, Res-cLSTM eliminates the need for extensive training datasets, significantly reducing training time compared to existing models that require a large number of training data.

The remainder of this article is structured as follows. The proposed Res-cLSTM is detailed in Section II. In Section III, experiments based on simulated and real clouds are conducted to demonstrate the effectiveness of the proposed Res-cLSTM. Section IV further discusses the effectiveness of Res-cLSTM, along with its potential capabilities and limitations. Section V concludes this article.

II. Methods

A. Overview of the Network Design of Res-cLSTM

In this article, we proposed Res-cLSTM for reconstructing cloud occlusion data in time-series cloudy images. The effective noncloud data in time-series cloudy images contain valuable information of temporal variation and also spatial structure. Res-cLSTM utilizes valid noncloud data from time-series cloudy images to train the models, thereby eliminating the need for extensive training datasets. As illustrated in Fig. 1, Res-cLSTM is an end-to-end model composed primarily of two modules: the multitemporal decoding convolutional LSTM module and the refining residual module. The convolutional LSTM module processes multitemporal data by inputting them sequentially and capturing the complex temporal relationships among them through the integration of long- and short-term memories. The refining residual module serves to further refine the feature maps for the final predictions. This research focuses on six bands of Landsat 8 OLI imagery: blue, green, red, near infrared (NIR), shortwave infrared 1 (SWIR 1), and shortwave infrared 2 (SWIR 2). Each Landsat 8 OLI image fed into the network includes these six bands. In Fig. 1, $\mathbf{T}_1$, $\mathbf{T}_2, \ldots, \mathbf{T}_{p-1}, \mathbf{T}_{p+1}, \ldots, \mathbf{T}_{n-1}, \mathbf{T}_n$ represent the input time-series auxiliary cloudy images, while $\mathbf{T}_p$ denotes the target cloudy image to be predicted. n is the number of images in the time-series, including the target cloudy image itself. Res-cLSTM reconstructs the cloudy images at successive time points, ultimately generating a series of reconstructed cloud-free images. The components and functionalities of each Res-cLSTM module are detailed in Sections II-B and II-C, respectively.

B. Multitemporal Decoding Module

Shi et al. [57] proposed the convolutional LSTM structure based on the fully connected layer-based LSTM and converted the precipitation prediction problem into the spatiotemporal series prediction problem from the perspective of machine learning. The original LSTM, based on fully connected layers, is a type of neural network characterized by memory capabilities and was designed primarily for processing 1-D time-series. However, it struggles to effectively capture spatial

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features. On this basis, the convolutional LSTM ingeniously replaces the fully connected layers with convolutional layers, enabling it to capture additional spatial structural information from the input data.

Given the satisfactory performance of convolutional LSTM in predicting time-series data, we proposed to utilize it to reconstruct multispectral time-series cloudy images. The first component of Res-cLSTM is the convolutional LSTM module, which consists of n-1 convolutional LSTM cells for the n-1 auxiliary time-series images. In this article, n was set to 5 (i.e., 4 auxiliary images were used). The time-series data are input into the convolutional LSTM module sequentially in chronological order. The number of channels in the hidden layer is configured to match the number of channels in the input layer, denoted as b. In this article, six bands of data were utilized (i.e., blue, green, red, NIR, SWIR1, and SWIR2) and, thus, b was set to six. Unlike LSTM, which employs fully connected layers, convolutional LSTM utilizes convolutional kernels, enhancing the model's ability to extract spatial information. This modification greatly increases the operational efficiency of the network and significantly reduces computational time. Moreover, by retaining long-term memory and integrating it with short-term memory for each new input, the convolutional LSTM module effectively captures temporal-varying relationships within the time-series data.

C. Refining Residual Module

The residual structure was first proposed in [58], which contains a short-circuit connection. Short-circuit connections can facilitate the effective reuse of features extracted from previous layers. By simplifying the optimization process, short-circuit connections often enable the network to converge more rapidly, resulting in more efficient training. Furthermore, short-circuit connections provide greater flexibility in adjusting the mapping between layers during the learning process, which improves the fitting capacity of network. Additionally, they can help mitigate the risk of overfitting, ultimately enhancing the model's generalization ability.

After processing through the convolutional LSTM module, the spatiotemporal information of the time-series data is learned preliminarily. However, we expect the network to further refine the results. Considering the above advantages of the residual structure, we designed a postprocessing module based on it, that is, the refining residual module. The output feature map is fed into the refining residual module to further decompose and learn the feature information of the target cloudy images. As illustrated in Fig. 1, the refining residual module comprises five convolutional layers, three of which are activated by the ReLU function. During the model training process, both the images and their corresponding masks are segmented into patches before being fed into the network. A loss function is formulated between the final output and the reference patches based on the L1 norm

$$\text{Loss} = \|(1-\mathbf{M}_p)\ \text{e}\ \left(f\left(\mathbf{T}_1, \mathbf{T}_2, \ldots, \mathbf{T}_{p-1}, \mathbf{T}_{p+1}, \ldots, \mathbf{T}_{n-1}, \mathbf{T}_n\right) - \mathbf{T}_p\right)\|_1 \tag{1}$$

where f represents the proposed Res-cLSTM network and $(\mathbf{T}_1, \mathbf{T}_2, \ldots, \mathbf{T}_{p-1}, \mathbf{T}_{p+1}, \ldots, \mathbf{T}_{n-1}, \mathbf{T}_n)$ and $\mathbf{T}_p$ represent the input time-series auxiliary images and the corresponding target cloudy image, respectively. $\mathbf{M}_p$ refers to the known cloud mask corresponding to the target cloudy image, presented as a binary matrix where 0 indicates noncloud regions and 1 signifies cloud regions. $\mathbf{1}$ is an all-one matrix with the same dimensions as $\mathbf{M}_p$. $\|\|_1$ signifies the L1 norm, and $\odot$ denotes the matrix dot product operation. The Res-cLSTM model was trained based on (1) to reconstruct sequentially the entire set of time-series cloudy images. In this article, Res-cLSTM underwent training for 120 epochs, starting with an initial learning rate of 0.01, which halved every 50 epochs.

D. Model Training and Predicting

Let $\mathbf{T}_p$ be the target cloudy image, $\mathbf{M}_p$ be its cloud mask, and $(\mathbf{T}_1, \mathbf{T}_2, \ldots, \mathbf{T}_{p-1}, \mathbf{T}_{p+1}, \ldots, \mathbf{T}_{n-1}, \mathbf{T}_n)$ be the corresponding time-series auxiliary data. The implementation of the proposed Res-cLSTM method is outlined as follows.

1) Training: The following conditions hold.

1) Partitioning $\mathbf{T}_p$, $\mathbf{M}_p$ and $(\mathbf{T}_1, \mathbf{T}_2, \ldots, \mathbf{T}_{p-1}, \mathbf{T}_{p+1}, \ldots, \mathbf{T}_{n-1}, \mathbf{T}_n)$ into $m \times m$ patches: $\mathbf{t}_j^{m\times m\times b}$, $\mathbf{m}_j^{m\times m\times 1}$, and $(\mathbf{t}_1^{m\times m\times b}, \mathbf{t}_2^{m\times m\times b}, \ldots, \mathbf{t}_{p-1}^{m\times m\times b}, \mathbf{t}_{p+1}^{m\times m\times b}, \ldots, \mathbf{t}_{n-1}^{m\times m\times b}, \mathbf{t}_n^{m\times m\times b})_j$. Here, b represents the number of spectral bands in each image, and $j = 1, 2, \ldots, J$, where J is the total number of patches resulting from the segmentation of each image.
2) Inputting the patches of time-series auxiliary data $(\mathbf{t}_1^{m\times m\times b}, \mathbf{t}_2^{m\times m\times b}, \ldots, \mathbf{t}_{p-1}^{m\times m\times b}, \mathbf{t}_{p+1}^{m\times m\times b}, \ldots, \mathbf{t}_{n-1}^{m\times m\times b}, \mathbf{t}_n^{m\times m\times b})_j$ into the Res-cLSTM network. First, the convolutional LSTM module processes the time-series data sequentially in chronological order, yielding an intermediate result $\mathbf{h}_j^{m\times m\times b}$. This intermediate output is then fed into the residual module to generate the final output $\mathbf{p}_j^{m\times m\times b}$ of Res-cLSTM. The loss is computed using $\mathbf{p}_j^{m\times m\times b}$, $\mathbf{t}_j^{m\times m\times b}$, and $\mathbf{m}_j^{m\times m\times 1}$ based on (1) to guide network training.

2) Predicting: Using the trained Res-cLSTM model f obtained from the training steps, the time-series data $(\mathbf{T}_1, \mathbf{T}_2, \ldots, \mathbf{T}_{p-1}, \mathbf{T}_{p+1}, \ldots, \mathbf{T}_{n-1}, \mathbf{T}_n)$ are fed to derive the prediction $\mathbf{P}_p$. Finally, the noncloud areas of $\mathbf{T}_p$ are concatenated with the reconstructed cloud area from $\mathbf{P}_p$ to yield the final prediction. The whole process is repeated for each cloudy image to produce results $\mathbf{P}_1, \ldots, \mathbf{P}_n$ in turn.

III. Experiments

A. Data and Experimental Design

This article focuses on reconstructing the cloud occlusion areas in the cloudy time-series images. A total of 25 Landsat 8 OLI images from five regions (five images for each region) were used for the experiments. Fig. 2 shows the geographical distribution of the five study regions. Fig. 3 shows all the images, each covered by 2000 × 2000 Landsat pixels, corresponding to an area of 60 × 60 km^2. As shown in Fig. 3, Regions 1–3 are based on simulated thick clouds, and the real land surface information under cloud are originally known. Therefore, the real surface reflectance values under the simulated clouds can be used for both quantitative and qualitative evaluation of the reconstructions. Regions 4 and

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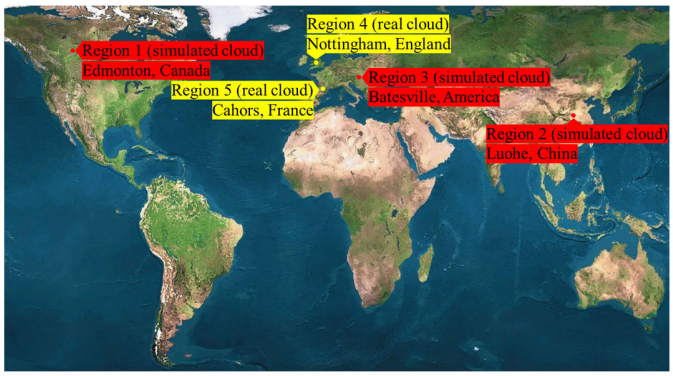


Fig. 2. Geographical locations of the study regions.

5 in Fig. 3 are real cloudy regions, where the accuracy of the reconstructions can be assessed visually only. Regions 1–5 all contain different forms of farmland, mountains, and water bodies, as well as urban areas. The types of ground objects are rich and diverse in Regions 1–5, especially the farmland with complex time changes, which poses a great challenge to the accurate reconstruction of the time-series images.

B. Comparison With Benchmark Methods

In this section, the proposed Res-cLSTM was compared with four benchmark methods, namely AE [31], STnet [41], DCR [32], and CR-former [50]. Among them, both Res-cLSTM and STnet used time-series data for training and prediction. The difference is that STnet inputs all the time-series auxiliary images simultaneously, while Res-cLSTM uses the convolutional LSTM module to feed them into the network in chronological order. Notably, AE, DCR, and CR-former require that the target cloud area corresponds to a completely cloudless region on the auxiliary images. Therefore, when there is an overlap between the cloud area of the auxiliary images and the target image, we stitched the predictions from multiple auxiliary images to obtain the final reconstruction for these three methods. Consequently, the training and prediction time costs increase proportionally with the number of spliced scenes. The reconstructions of one image (with two subareas zoomed-in view on the right) from each region (i.e., L11, L25, and L31) are shown in Fig. 4. It can be seen that the five methods can generally produce predictions with good consistency, especially in Region 2. The more detailed reconstructions of each method can be seen from the zoomed-in view subareas in each region, whose locations are marked by yellow and black boxes in Fig. 4. For L11, STnet, and Res-cLSTM predicted the black objects in Subarea 1 and the green objects in Subarea 2 more accurately than AE and DCR that use only single auxiliary images. In the two subareas of L25, it can also be seen that Res-cLSTM and STnet reconstructed the distribution of red objects (i.e., vegetation) more accurately. This may be because the cloud region in the temporally closest auxiliary image overlaps with the target cloud region. As a result, AE and DCR must rely on a temporally further auxiliary image that does not contain overlapping cloud regions, significantly diminishing the effectiveness of the auxiliary image. In contrast, the method based on time-series auxiliary data can utilize temporally closer cloudy auxiliary images, which is conducive to obtaining more accurate predictions. For CR-former, the predictions exhibit tonal anomalies in the two subareas of L11, although it accurately predicts the green features in the second subarea. Moreover, CR-former fails to restore the red features in both subareas of L25. In contrast, for L31, the cloud region of the temporally closest auxiliary image does not overlap with its cloud region, and AE, DCR, and CR-former can reconstruct the cloud-contaminated information more satisfactorily than STnet in this case. For L31, Res-cLSTM can reconstruct the texture details of ground objects most accurately and its prediction is the closest to the reference image in tone. Overall, among the five methods, the proposed Res-cLSTM consistently produces predictions that are the closest in tone and texture to the reference images.

In this article, the root-mean-square error (RMSE), correlation coefficient (CC), and universal image quality index (UIQI) were used to evaluate quantitatively the prediction of each method. The average accuracy of each prediction in six bands is shown in Fig. 5, which shows that the proposed Res-cLSTM generally produces the most accurate predictions. For example, for L11, the average RMSE of Res-cLSTM prediction is 0.0009, 0.0063, 0.0033, and 0.0028 smaller than that of the AE, STnet, DCR, and CR-former predictions, respectively; the average CC is 0.0170, 0.0878, 0.0469, and 0.0461 larger; and the average UIQI is 0.0182, 0.1026, 0.0488, and 0.0506 larger. In most cases, the AE predictions are second only to that of the Res-cLSTM predictions. Taking Region 2 as an example, the average CCs of the Res-cLSTM predictions of five temporal images were 0.0276, 0.0678, 0.0239, 0.0109, and 0.0290 greater than that of AE predictions. The results of quantitative evaluation show that Res-cLSTM can stably produce the most accurate predictions.

C. Influence of Thick Cloud in the Auxiliary Time-Series

Cloud occlusion on the auxiliary image will reduce the available valid data and may affect the accuracies of the final predictions. This section aims to test the effect of cloud data in the auxiliary image on the performance of each method. Specifically, we reconstructed the same target cloudy image using the simulated cloudy auxiliary time-series images (the same as in Section III-B) and their corresponding original cloud-free data, respectively, and compared the prediction accuracies in the two cases.

Fig. 6 shows the average CC of all predictions across six bands, from which it can be seen that the cloud-contaminated data in the auxiliary time-series images generally reduced prediction accuracy noticeably. For STnet, the clouds significantly affect the performances in most cases (e.g., the predictions of L11, L13, and L23 in Fig. 6). The reason may be that STnet input time-series images simultaneously for model training and prediction, which fails to fully utilize the temporal variation information. In contrast, Res-cLSTM processes the time-series data sequentially in chronological order and is capable of transmitting the valid information of the previous image to the next image. The mode of combining long- and short-term memory enables Res-cLSTM

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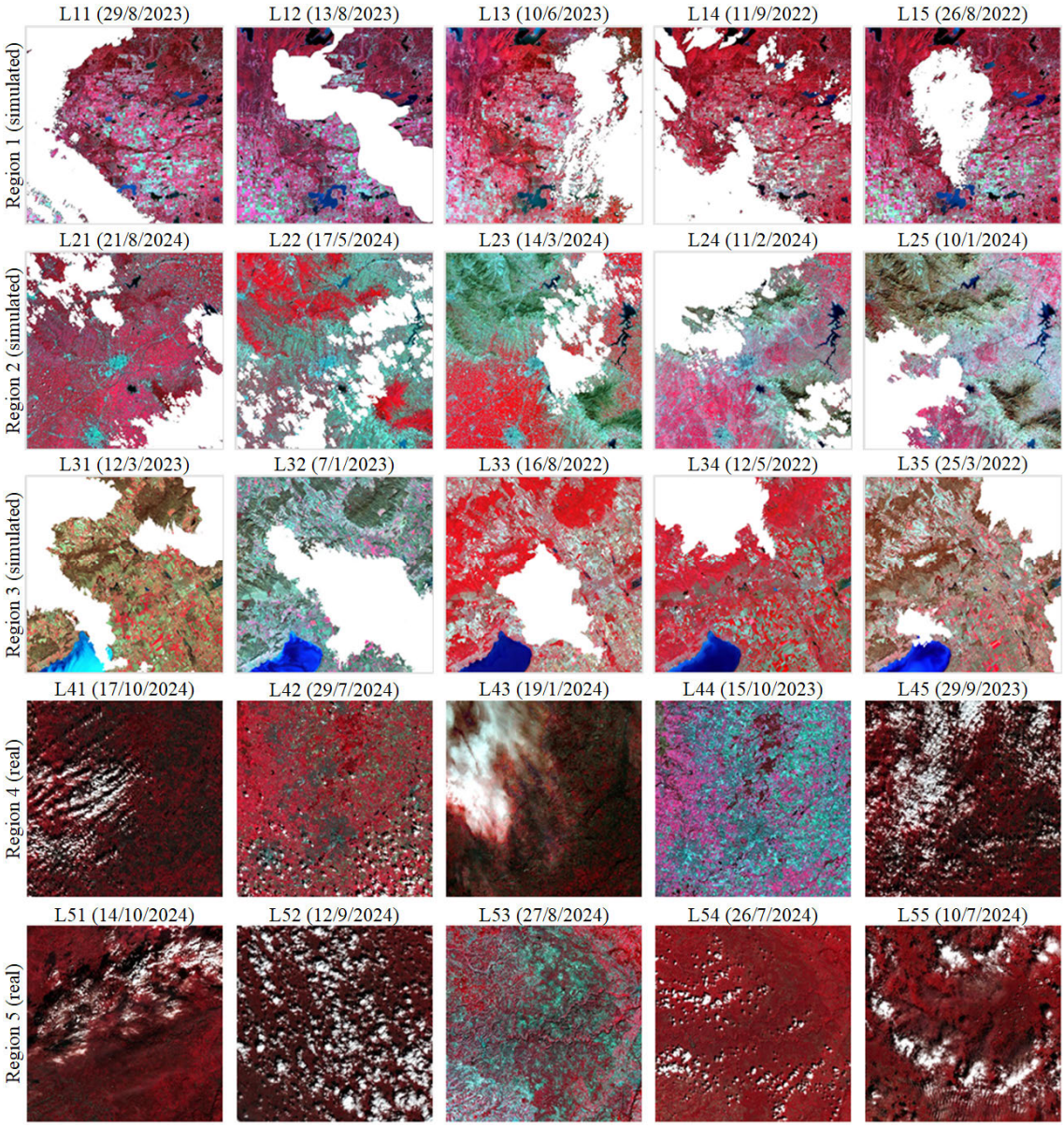


Fig. 3. Simulated and real cloudy time-series images (each with a size of 2000 × 2000 Landsat pixels) in Regions 1–5. From right to left, the images are arranged in chronological order. Above each image is its number and acquisition date (day/month/year).

to effectively complement the effective information in the noncloud areas of different auxiliary images. Thus, Res-cLSTM is relatively less affected by time-series cloud contamination than STnet. In Region 1, the accuracies of DCR predictions are significantly reduced when cloudy time-series images are used in most cases. This may be due to the drastic temporal changes in this region, and the cloud region in the temporally closest auxiliary image overlaps with the target cloud region, resulting in the use of a temporally further auxiliary image. The prediction accuracy of CR-former also shows obvious decreases in some results, such as L13 and L33. In contrast, AE shows relatively stable performance, especially in Region 3, since the cloud region of the temporally closest auxiliary image does not overlap with the target cloud region. Meanwhile, Res-cLSTM is also relatively less affected by clouds in the time-series images and consistently achieves the greatest performance. This leads to the conclusion that Res-cLSTM is more robust to cloud contamination in auxiliary time-series images than the four benchmark methods.

D. Influence of Cloud Overlap Rate

For cloudy time-series images, the accuracy of predictions may be affected when their cloud area overlaps with the target cloud area. In the cloud overlap area, some cloud removal methods may even fail completely (e.g., the method based on single auxiliary image). The purpose of this section is to explore the effect of the overlap rate between the auxiliary image cloud region and the target cloud region. As shown in Fig. 7, different cloud overlap rates were obtained by simultaneously moving a fixed-size simulated cloud region in the direction indicated by the blue arrows. Among them, masks

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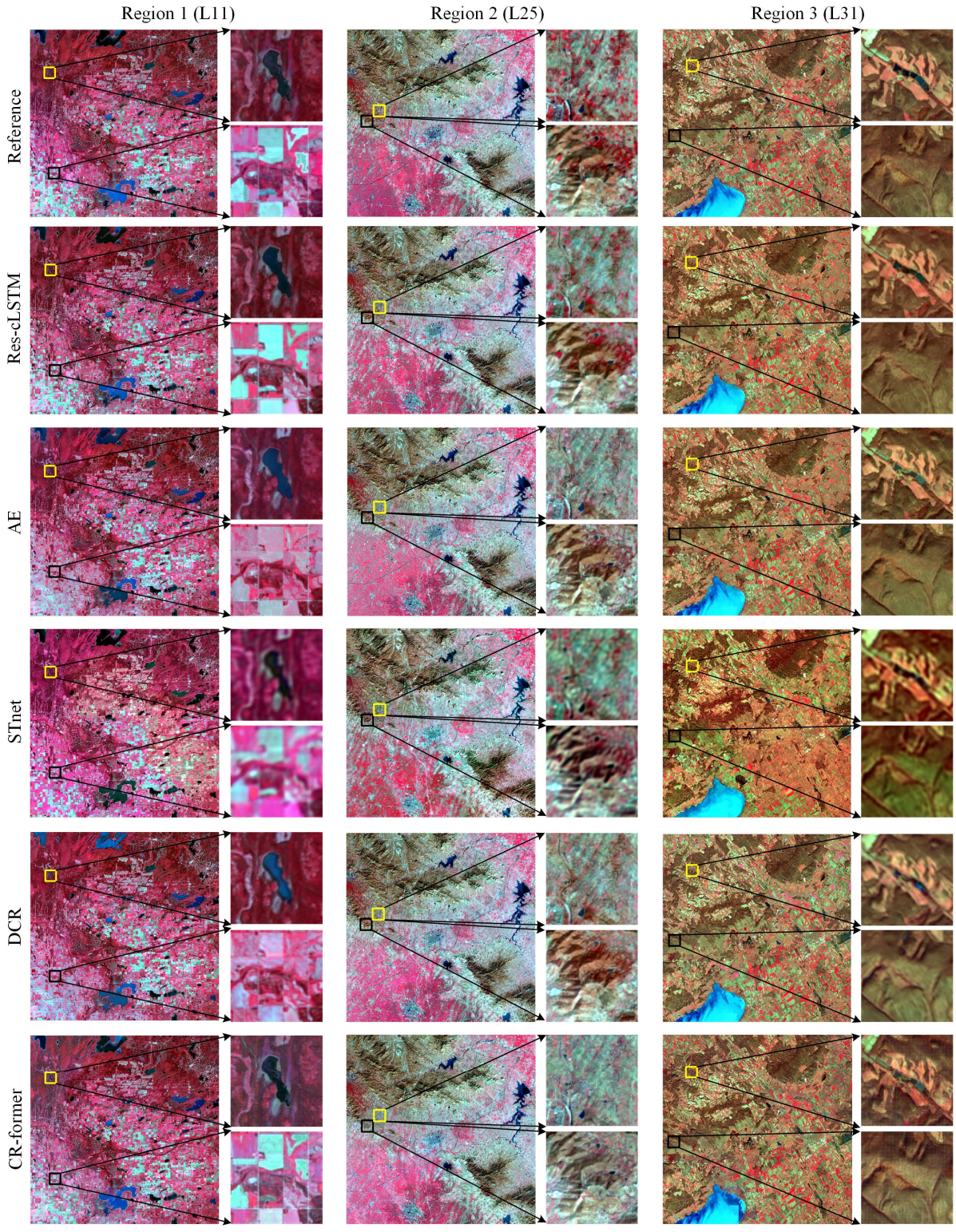


Fig. 4. Cloud removal results of the five methods in Regions 1–3 (with two subareas shown on the right; NIR, red, and green bands as RGB).

1–4 are the cloud masks (500 × 500 Landsat pixels) of the four time-series images, and mask target is the cloud mask (1000 × 1000 Landsat pixels) of the target cloudy image. For methods using single auxiliary images (i.e., AE and DCR), the auxiliary images acquired on two different dates were used to reconstruct the same target cloudy image, and then, the two results were spliced to obtain the final predictions, to avoid complete failure zones. In this section, the cloud overlap rates were calculated based on all auxiliary time-series images. For example, when the overlap rate is 80%, the proportion of cloud overlap area in each auxiliary image is 20% of the target cloud area. For DCR and AE, they only use the predictions of two

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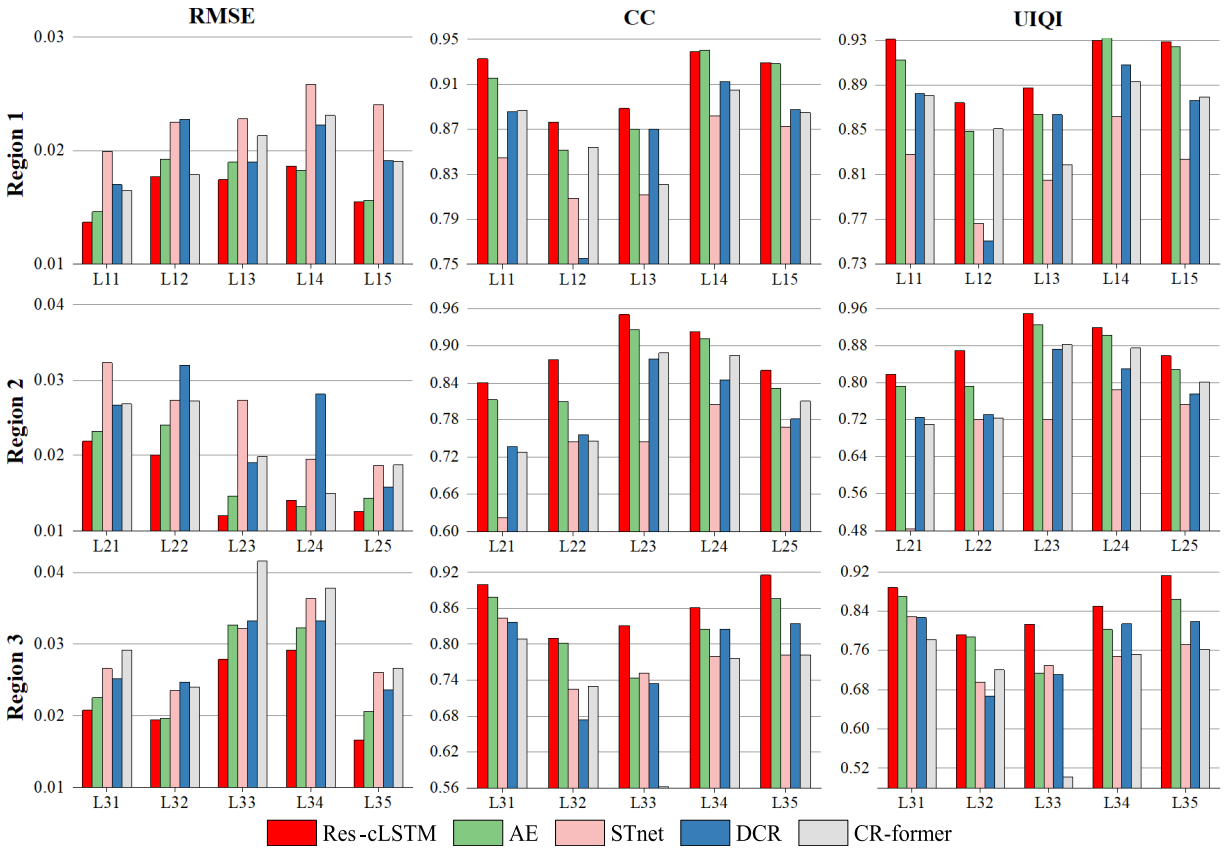


Fig. 5. Accuracies of the five cloud removal methods in Regions 1–3.

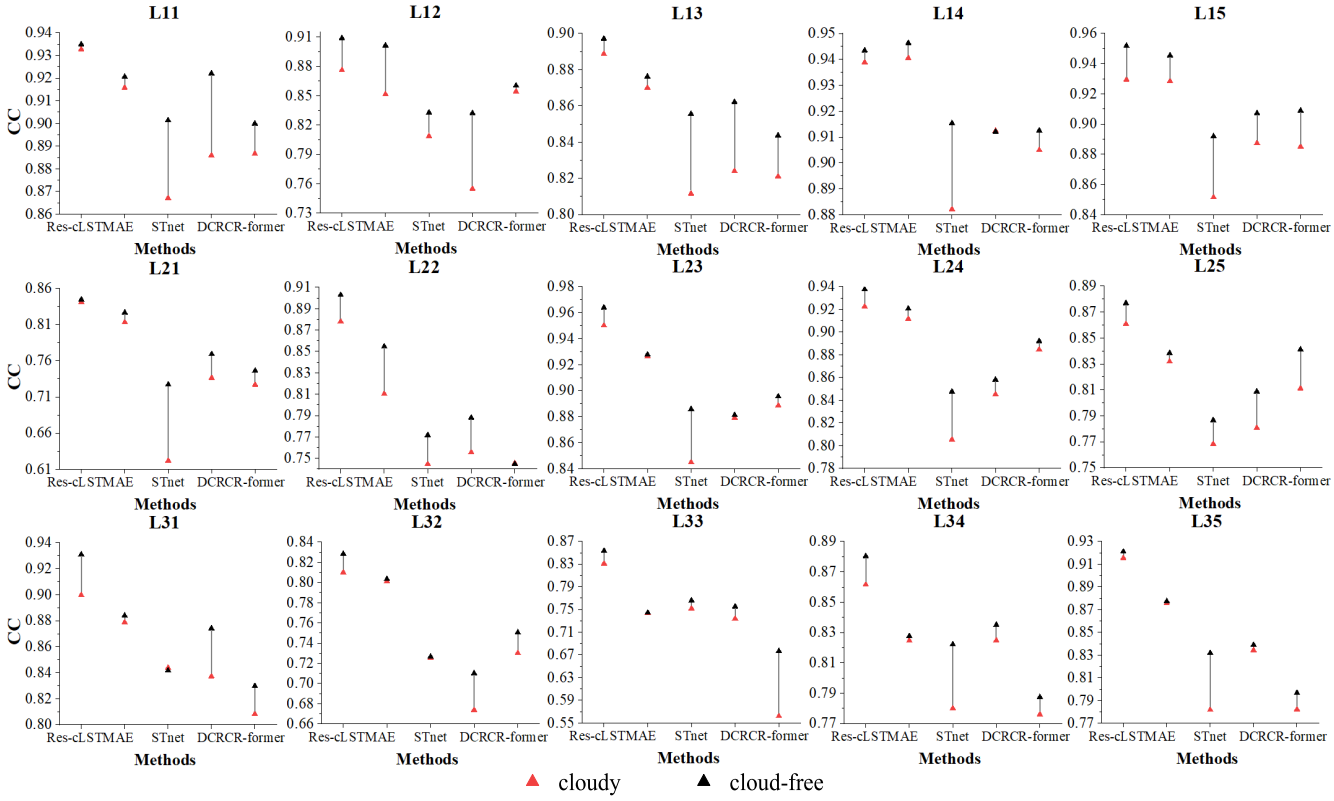


Fig. 6. CCs (averages of the six bands) of the five cloud removal methods using cloudy or cloud-free time-series auxiliary images in Regions 1–3.

auxiliary images to concatenate, and the cloud overlap rate on them is only 40%.

The results for the first image in each region (i.e., L11, L21, and L31) are shown in Fig. 8. It can be seen that the

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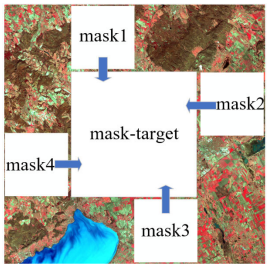


Fig. 7. Schematic of cloud mask position under different cloud overlap rates. The fixed-size masks 1–4 (corresponding to the four auxiliary temporal images) move in the direction indicated by the blue arrow to obtain different overlap rates with the mask of target cloudy image (mask target).

prediction accuracies of the five methods generally decrease as the cloud overlap rate increases. Among them, the accuracies of the STnet predictions fluctuate greatly in all three regions and decrease significantly in general, which may be because all auxiliary time-series images are inputted simultaneously to reconstruct the target cloud region. The influence of the cloud overlap areas on the DCR predictions in Regions 1 and 3 is relatively small, which may be due to the relatively smooth temporal changes in the two regions. However, in Region 2, the obvious decreases in the DCR prediction accuracies may be due to the large land cover changes in the temporally further auxiliary images. In contrast, AE and Res-cLSTM are less affected by cloud overlap areas, and the changes in the prediction accuracies are relatively stable in the three regions. Moreover, in almost all cases, Res-cLSTM performs more satisfactorily than AE. Thus, it can be concluded that Res-cLSTM can produce more accurate predictions consistently when there is a varying degree of cloud overlap between the auxiliary images and the target cloudy image.

E. Impact of Thin Cloud Omission

The previous experiments were all based on simulated cloud masks, which were known perfectly. Practically, cloud detection results commonly contain uncertainties, resulting in omission of thin clouds that are treated as cloud-free data during model training and prediction. In this section, we tested the influence of undetected thin cloud on the methods. Specifically, we used the method proposed by Guo et al. [59] to generate simulated thin clouds by mimicking the real situation of light penetrating clouds. Fig. 9 illustrates the distribution of simulated thin and thick clouds on L34, with the yellow coil indicating the simulated thin clouds. We incorporated varying proportions of these simulated thin clouds into 15 cloudy images across Regions 1–3 (as outlined in Section III-B). The percentages of thin clouds in total cloud cover range from 9.23% to 43.33%, as detailed in Table I.

Fig. 10 presents the accuracies of predictions with and without thin cloud omission for each method. The accuracies present a general decrease across all methods due to thin cloud omission, particularly for STnet, DCR, and CR-former. In contrast, AE and Res-LSTM demonstrated greater robustness against the effects of thin cloud omission. Overall, Res-cLSTM consistently achieved the greatest accuracy, maintaining a relatively low decrease in performance. This suggests that Res-cLSTM offers considerable promise for practical applications, remaining effective even in the presence of undetected thin clouds.

TABLE I
RATIO OF THIN CLOUDS TO THE TOTAL CLOUD COVER PERCENTAGE (X REPRESENTS THE REGION NUMBER)

	Lx1	Lx2	Lx3	Lx4	Lx5
Region 1	9.63%	13.52%	12.69%	11.93%	9.23%
Region 2	17.67%	43.33%	39.32%	25.43%	29.92%
Region 3	9.84%	10.42%	13.69%	20.73%	14.75%

F. Network Convergence Rate

The convergence rate is a crucial metric for deep learning networks, as it influences directly their effectiveness in practical applications. In this section, we evaluated the convergence rates of the five methods by analyzing the increase in accuracy over time. The proposed Res-cLSTM was implemented using Python 3.7 on a personal computer equipped with an Intel Core i9@3.70 GHz and an NVIDIA GeForce RTX 3070.

The results are presented in Fig. 11. Each curve begins with the accuracy achieved after training each network for one epoch and ends at the point where the increase in accuracy for each method significantly slows down. Notably, the prediction accuracy of AE reaches 95% of its final level after just one epoch. This can be attributed to its reliance on multiband information from single pixels for model training and prediction, resulting in a relatively straightforward learning relationship and a limited maximum attainable accuracy. In contrast, DCR and CR-former feature a deeper network structure, which restricts its learning rate and prolongs training time per epoch. Both Res-cLSTM and STnet demonstrate comparable and rapid operating speeds (approximately 5–6 s per epoch), with their prediction accuracies increasing quickly and significantly at the outset. However, Res-cLSTM ultimately achieves substantially greater prediction accuracy than STnet. These results indicate that Res-cLSTM exhibits a faster convergence rate and is capable of delivering more accurate predictions in a shorter timeframe. Consequently, Res-cLSTM shows considerable potential for development and real-world applications, with its rapid convergence facilitating real-time generation of up-to-date predictions.

G. Experiments on Real Clouds

In this section, real cloudy images in Regions 4 and 5 were employed to evaluate the performance of the five cloud removal methods. As illustrated in Fig. 3, L44 and L53 are cloud-free images, while the remaining images contain clouds. The predictions of the five methods are presented in Fig. 12. Notably, when the cloud-contaminated area of the target image is excessively large (e.g., in L43 and L55, where both have cloud cover exceeding 70%), the DCR and CR-former predictions exhibit significant tonal anomalies. This phenomenon may be attributed to the limited availability of

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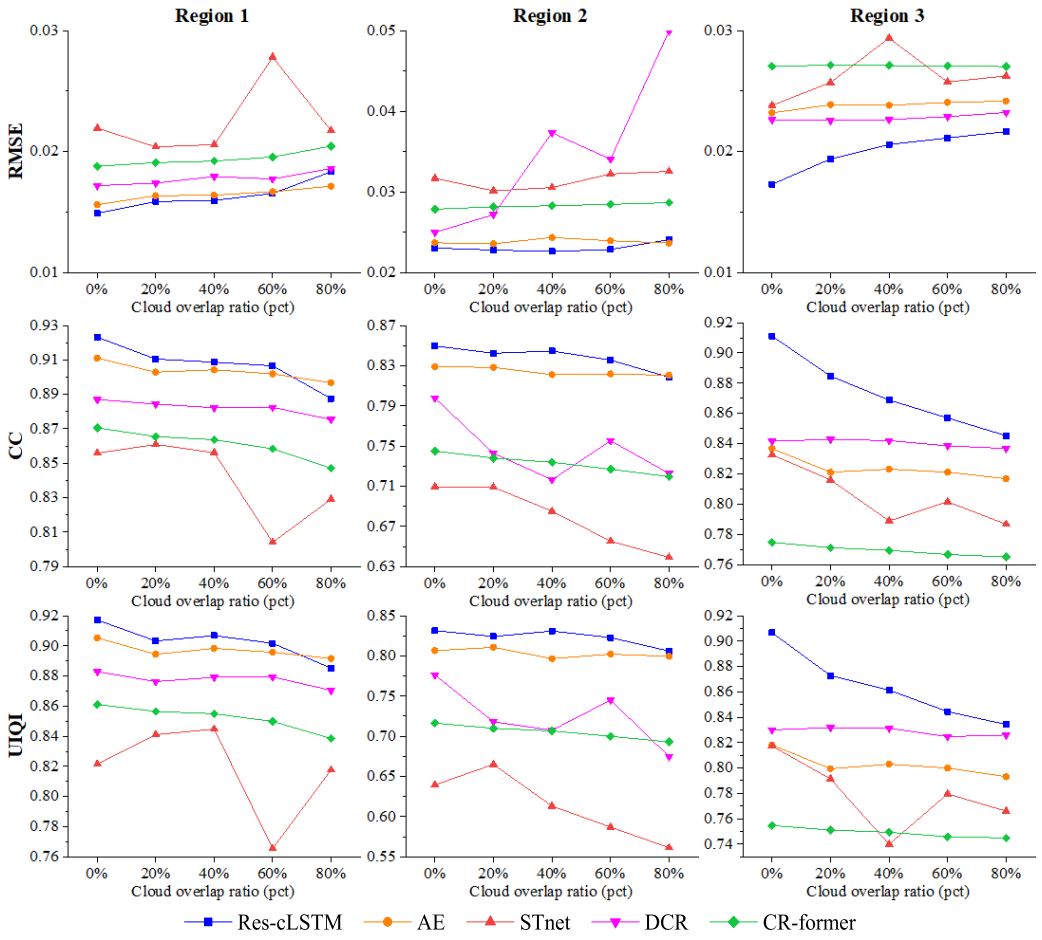


Fig. 8. Accuracies (averages of the six bands) of the five cloud removal methods with different cloud overlap ratios.

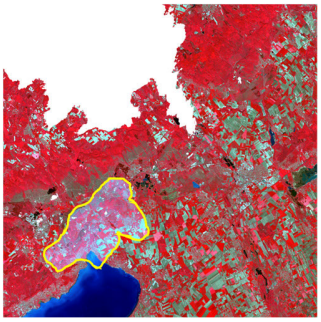


Fig. 9. Distribution of the simulated thick and thin clouds (taking L34 as an example). The yellow line delineates the simulated thin cloud.

training data, which hampers the effective training of the DCR models. In contrast, predictions generated by Res-cLSTM, AE, and STnet show satisfactory tonal consistency. Fig. 13 presents a detailed view of a subarea from each reconstructed time-series, with their locations indicated by red boxes on the cloud masks in Fig. 12. The subareas for L42 and L54 are cloud-free. It is evident that predictions produced by STnet are somewhat blurry. Compared with STnet and DCR, Res-cLSTM and AE effectively recover more texture details. In addition, Res-cLSTM can recover effective temporal variation, as characterized by the reconstructed time-series.

IV. Discussion

A. Advantages of the Proposed Res-cLSTM

In this article, we proposed a deep network model called Res-cLSTM for reconstructing missing data in time-series cloudy images. The effectiveness of Res-cLSTM was demonstrated in Section III, where it achieves more accurate cloudy time-series data reconstruction results compared to several benchmark methods. Unlike traditional methods that require cloud-free auxiliary images, Res-cLSTM effectively utilizes the available noncloud data in cloudy time-series images. Moreover, while most existing methods focus on reconstructing single cloudy images, Res-cLSTM excels at efficiently reconstructing time-series cloudy images jointly, enabling more effective monitoring of dynamic land surface changes and offering greater practical significance. Furthermore, Res-cLSTM is a relatively lightweight network that can be trained solely using noncloud data from time-series cloudy images, eliminating the need to gather extensive training datasets globally and greatly simplifying data collection. Additionally, compared with the three benchmark methods, Res-cLSTM exhibits faster convergence and provides more accurate predictions in a shorter time frame, showcasing significant practical application potential, particularly in scenarios that require real-time processing.

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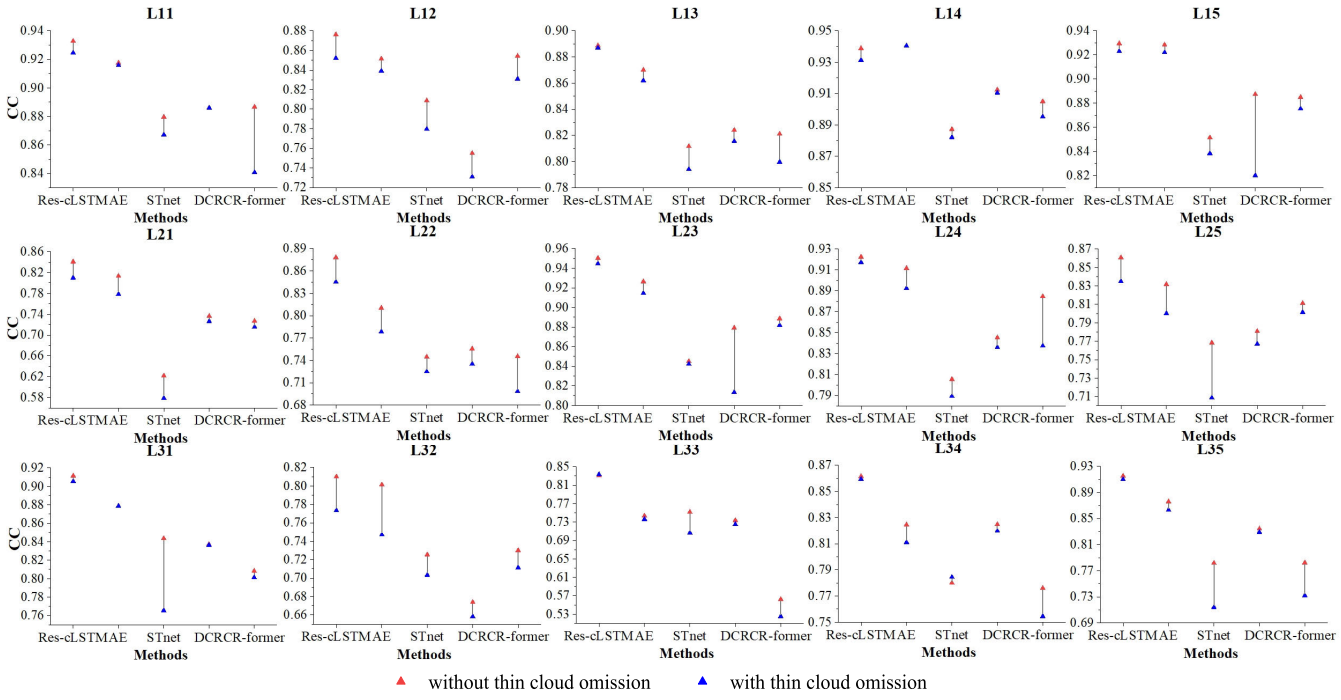


Fig. 10. CCs (averages of the six bands) of the five cloud removal methods based on images with or without thin cloud omission in Regions 1–3.

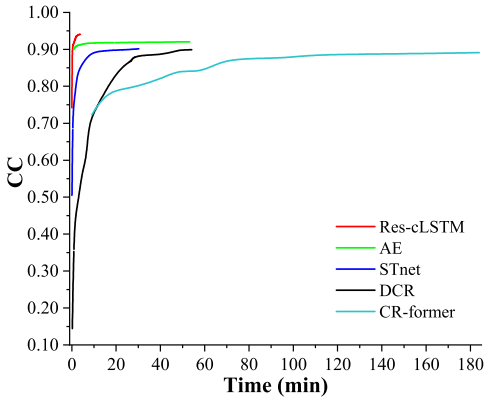


Fig. 11. Changes in CCs (averages of the six bands) of the five cloud removal methods over computing time.

B. Generalization Ability of Res-cLSTM

In this article, the Res-cLSTM network leverages a known cloud mask to exclude cloud region data from the input cloudy images during the training process. Notably, Res-cLSTM has potential applicability in handling other types of missing data, such as quantitative remote sensing data. Employing the convolutional LSTM originally designed for precipitation prediction, Res-cLSTM is expected to reconstruct datasets with clear temporal relationships, including land surface temperature (LST), fractional vegetation cover (FVC), and nighttime lights. Given that Res-cLSTM was applied to multispectral time-series data in this research, the rich spectral information may also be a major reason to promote accurate predictions. For quantitative remote sensing data that may contain limited spectral information, integrating auxiliary data from other sources could be a more effective approach. For example, when reconstructing missing LST data, the incorporation of elevation and geographical information (such as latitude and longitude) could provide valuable context. However, how to optimally leverage this effective information requires further experimental investigation. Additionally, Res-cLSTM does not impose stringent requirements on data resolution, allowing for flexibility in its application. By utilizing the adaptable nature of convolution operations across different spatial scales, the Res-cLSTM model can be readily developed and employed to reconstruct data at various spatial resolutions.

C. Differentiated Removal of Thick and Thin Clouds

The proposed Res-cLSTM removes clouds from cloudy time-series images using known cloud masks for both simulated and real cloudy images, without differentiating between thick and thin clouds. However, thin clouds often allow some underlying land surface information to be visible, and treating them the same as thick clouds does not fully leverage this valuable information. In practical applications, cloud masks are generated using specialized cloud detection methods [60], [61]. However, thick and thin clouds frequently coexist and can be difficult to distinguish. Given the powerful learning capabilities of neural networks, there is potential to develop models that can differentiate between thin and thick clouds during the removal process. For example, by utilizing the strong learning capabilities of deep networks, specific loss functions can be designed to guide the automatic learning of thick and thin cloud features within the network, enabling targeted reconstruction of both types simultaneously.

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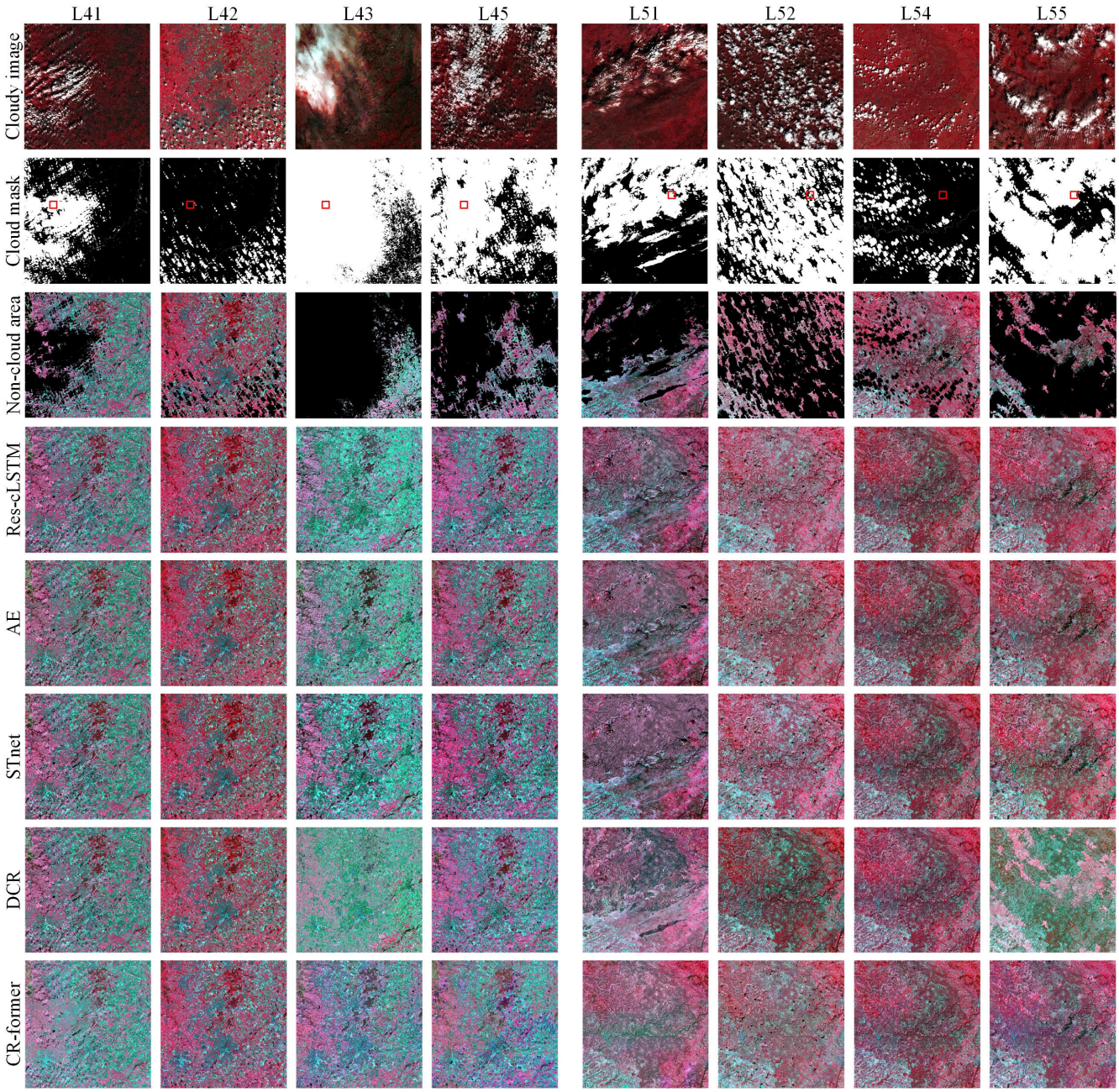


Fig. 12. Cloud removal results of the five methods in Regions 4 and 5 (NIR, red, and green bands as RGB).

D. Computational Cost

Fig. 14 illustrates the prediction accuracy and corresponding training time of the Res-cLSTM model for a target cloudy image with varying numbers of auxiliary images. The target cloudy images for the three regions are L11, L21, and L31. In Fig. 14, CC1–CC3 represent the average CC of the reconstructions for L11, L21, and L31 across six bands, while Time1–Time3 indicate the corresponding time consumption. Notably, when the number of images exceeds 4, the time consumption increases sharply, but the prediction accuracy decreases in Regions 2 and 3. In Region 1, the differences in prediction accuracy with varying numbers of auxiliary images are slight, likely due to the small temporal variations amongst the time-series images. Consequently, four auxiliary images were used in this article.

E. Uncertainty in Res-cLSTM

The proposed Res-cLSTM reconstructs time-series cloudy images by modeling the temporal relationships between them. In practice, the acquisition dates of time-series data are random. In regions consistently covered by clouds, Res-cLSTM is constrained by the scarcity of auxiliary information. Moreover, when adjacent images are captured in different seasons, significant changes in land surface characteristics may occur, increasing the risk of model failure. Given that Res-cLSTM is flexible in the number of time-series data it

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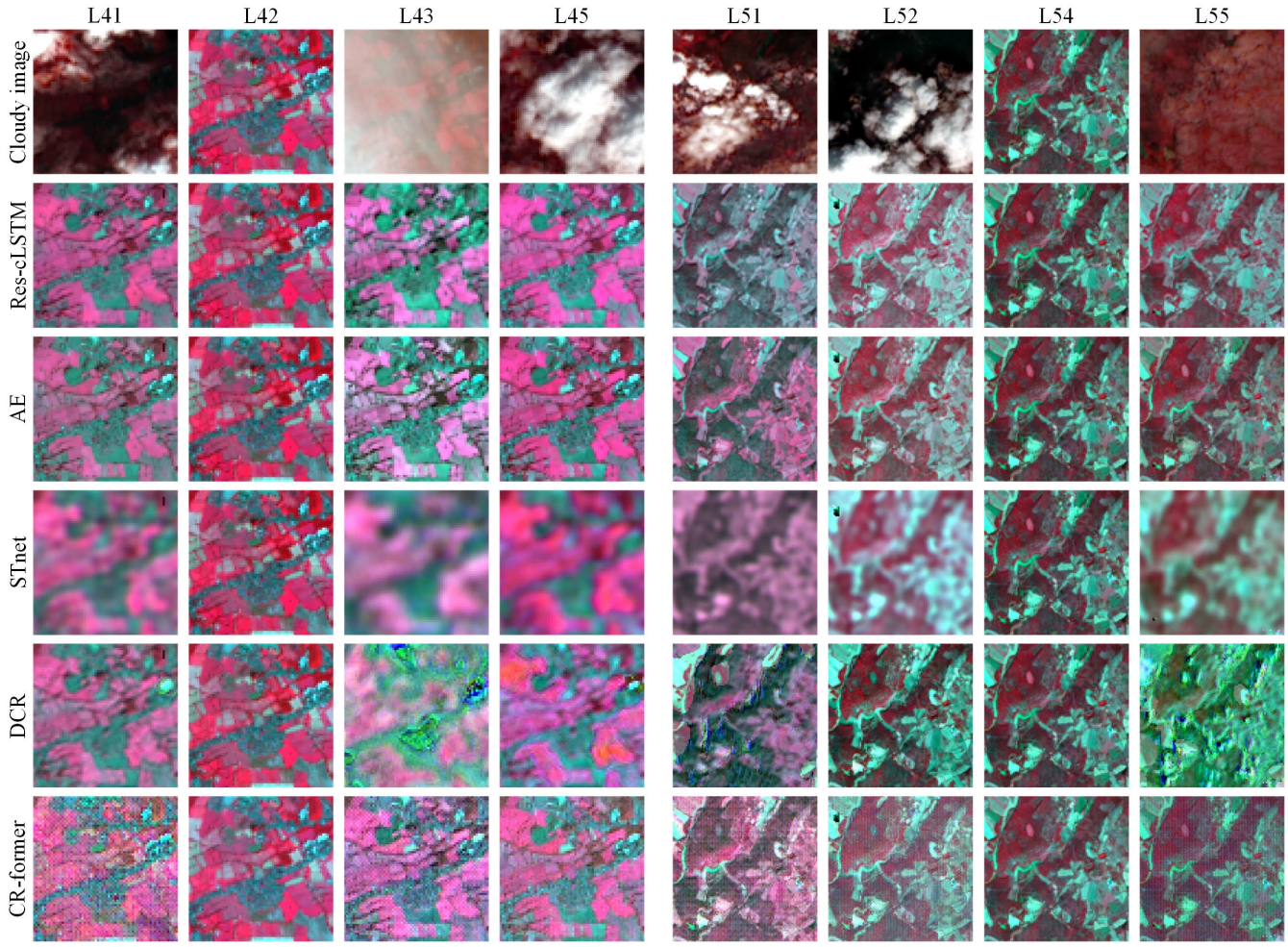


Fig. 13. Subareas of the predictions in Fig. 10 (NIR, red, and green bands as RGB).

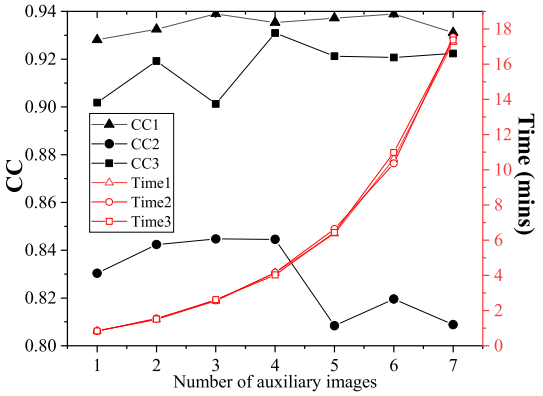


Fig. 14. Computational cost and accuracy of Res-cLSTM under different numbers of auxiliary images in Regions 1–3.

utilizes, future implementations could incorporate embedding modules or formulae to assess the severity of the temporal changes. This enhancement may increase the accuracy of predictions for other time-series images by eliminating those with large temporal changes. Subsequently, the reconstructed time-series images could be used to target and reconstruct the eliminated cloudy images that exhibit large temporal changes. Moreover, since Res-cLSTM reconstructs time-series cloudy images based on known cloud masks, the accuracy of its outputs may be influenced by the precision of the mask in practical applications. Integrating interactive modules capable of learning and refining the cloud masks during Res-cLSTM training could provide an alternative approach to enhance predictions while reducing reliance on known cloud masks. Additionally, auxiliary images from other sensors (such as SAR) could be integrated to further optimize prediction.

F. Comparison With Nondeep Learning-Based Methods

In this section, we compared the proposed Res-cLSTM with three traditional (i.e., nondeep learning-based) cloud removal methods that utilize time-series auxiliary data: 1) a modified neighborhood similar pixel interpolator approach for removing thick clouds based on Landsat time-series auxiliary data, abbreviated as multi-NSPI [44]; 2) a missing observation prediction based on spectral–temporal metrics (MOPSTM) gap-filling method [45]; and 3) an algorithm for generating synthetic Landsat images based on all available Landsat data, referred to as AGSL [43]. We evaluated the performance of each method using both cloudy and cloud-free auxiliary images, with the results presented in Tables II and III.

As shown in Table II, the predictions produced by Res-cLSTM exhibit obviously larger accuracy compared to the

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TABLE II

ACCURACIES OF THE FOUR CLOUD REMOVAL METHODS (RES-CLSTM AND NONDEEP LEARNING-BASED) PRODUCED USING CLOUD-FREE AUXILIARY TIME-SERIES DATA (REGION 1 AS AN EXAMPLE; THE VALUE IN BOLD MEANS THE MOST ACCURATE RESULT IN EACH CASE)

		Res-cLSTM	Multi-NSPI	MOPSTM	AGSL
L11	RMSE	**0.0131**	0.0162	0.0420	0.0218
	CC	**0.9349**	0.8975	0.8702	0.8510
	UIQI	**0.9337**	0.8959	0.8131	0.8442
L12	RMSE	**0.0149**	0.0179	0.0557	0.0341
	CC	**0.9089**	0.8587	0.7116	0.7553
	UIQI	**0.9076**	0.8556	0.5999	0.7073
L13	RMSE	**0.0170**	0.0199	0.0294	0.0392
	CC	**0.8971**	0.8554	0.8137	0.8076
	UIQI	**0.8958**	0.8504	0.7855	0.7352
L14	RMSE	**0.0176**	0.0229	0.0388	0.0278
	CC	**0.9435**	0.9039	0.8921	0.8869
	UIQI	**0.9408**	0.9006	0.8255	0.8465
L15	RMSE	**0.0162**	0.0174	0.0381	0.0300
	CC	**0.9230**	0.9131	0.8604	0.8514
	UIQI	**0.9216**	0.9100	0.8279	0.8191

TABLE III

ACCURACIES OF THE FOUR CLOUD REMOVAL METHODS (RES-CLSTM AND NONDEEP LEARNING-BASED) PRODUCED USING CLOUDY AUXILIARY TIME-SERIES DATA (REGION 1 AS AN EXAMPLE; THE VALUE IN BOLD MEANS THE MOST ACCURATE RESULT IN EACH CASE)

		Res-cLSTM	Multi-NSPI	MOPSTM	AGSL
L11	RMSE	**0.0137**	0.0243	0.0244	0.0245
	CC	**0.9328**	0.8264	0.8100	0.8464
	UIQI	**0.9309**	0.8142	0.7772	0.8157
L12	RMSE	**0.0177**	0.0212	0.0297	0.0441
	CC	**0.8762**	0.7888	0.7008	0.4832
	UIQI	**0.8742**	0.7841	0.6676	0.4716
L13	RMSE	**0.0174**	0.0256	0.0384	0.0415
	CC	**0.8887**	0.7828	0.8227	0.7603
	UIQI	**0.8876**	0.7731	0.7560	0.6984
L14	RMSE	**0.0186**	0.0290	0.0311	0.0300
	CC	**0.9389**	0.8528	0.8874	0.8725
	UIQI	**0.9300**	0.8408	0.7487	0.8400
L15	RMSE	**0.0155**	0.0222	0.0228	0.0283
	CC	**0.9295**	0.8585	0.8377	0.8375
	UIQI	**0.9287**	0.8514	0.8131	0.8033

three nondeep learning methods. This discrepancy may stem from the reliance of multi-NSPI and MOPSTM on the spatial neighborhood information of the target cloud pixels, which poses challenges for reconstructing large-scale missing areas, such as the extensive cloud occlusion present in the 2000 × 2000 Landsat pixel experimental region here. Furthermore, our experiments utilized only four auxiliary temporal images, which falls short of the minimum recommended by AGSL (at least eight scenes). This limitation could be a significant factor affecting the accuracy of AGSL. Additionally, traditional approaches often emphasize establishing explicit relationships based on spatially and temporally adjacent information, which may lack sufficient descriptive power for addressing the complex mapping relationships in real-world scenarios.

Moreover, as indicated in Table III, the prediction accuracy of the three conventional methods declines sharply when the time-series auxiliary images are affected by clouds. This decline is primarily due to the further reduction in available effective data, which exacerbates the limitations mentioned above. In contrast, Res-cLSTM consistently delivers more accurate predictions, regardless of whether the time-series data are cloudy or cloud-free, demonstrating superior robustness against cloud occlusion in the auxiliary data.

V. CONCLUSION

In this article, a deep learning method (i.e., Res-cLSTM) was proposed for reconstructing cloudy time-series images. Res-cLSTM leverages effective data from the noncloud regions of cloudy time-series images for reconstruction. First, Res-cLSTM employs a convolutional LSTM to sequentially encode the cloudy time-series images in their temporal order, effectively capturing the complex temporal changes through the integrated long- and short-term memory flow. Then, Res-cLSTM utilizes a refined residual module to further decode the feature map and generate the final predictions. Experiments were conducted using Landsat 8 OLI time-series data across five different regions, yielding the following conclusion.

1) Compared to the four benchmark methods, Res-cLSTM produced more accurate predictions that are similar to the reference images, confirming its effectiveness for reconstructing cloudy time-series images.
2) Res-cLSTM can achieve the most accurate predictions even when the time-series auxiliary images contain clouds, showing resilience against the influence of cloud data in the auxiliary time-series.
3) Under varying degrees of overlap (up to 80%) between the cloud regions in the auxiliary time-series images and the target cloud regions, Res-cLSTM consistently produced predictions with greater accuracies.
4) Res-cLSTM demonstrates robustness against thin cloud omission and shows greater potential for practical implementations.
5) Res-cLSTM is a relatively lightweight network boasting high computational efficiency and a convergence speed that significantly exceeds that of the four benchmark methods, showing significant potential for real-time processing in practical applications.

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